THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 3: **A Story of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Roman numeral XLIX stands for: 1

(a) 50 **(b)** 59 **(c)** 39 **(d)** 49

2. The Roman numeral MCMXCIV stands for: 1

(a) 1994 **(b)** 2004 **(c)** 1984 **(d)** 1995

3. 99 in Roman numerals is: 1

(a) XCVIII **(b)** CIX **(c)** XCIX **(d)** C

4. The decimal number 30 written in base 2 is: 1

(a) 100000 **(b)** 11110 **(c)** 11101 **(d)** 11111

5. The decimal number 44 written in base 6 is: 1

(a) 112 **(b)** 113 **(c)** 111 **(d)** 114

6. The value of 212 in base 3, in the decimal system, is: 1

(a) 25 **(b)** 23 **(c)** 24 **(d)** 22

7. Our present system of writing numbers with the digits 0 to 9 originated in: 1

(a) Greece **(b)** India **(c)** Egypt **(d)** Rome

8. The Italian mathematician who argued for adopting Indian numerals in Europe around 1200 was: 1

(a) Fibonacci **(b)** Laplace **(c)** Al-Kindi **(d)** Brahmagupta

9. In Roman numerals, a smaller symbol written before a larger one is: 1

(a) ignored **(b)** multiplied **(c)** added **(d)** subtracted

10. The value of the digit 4 in 5400 (decimal) is: 1

(a) 4 ones **(b)** 4 hundreds, i.e. 400

(c) 4 tens **(d)** 4 thousands

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is one-to-one mapping? Give an example from counting cattle. 2

12. Write 2024 in Roman numerals. 2

13. What do we mean by the base of a number system? 2

14. Why is the Roman system not suitable for large calculations? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Write 1444 in Roman numerals and explain the grouping. 3

16. Write 399 in Roman numerals and explain the grouping. 3

17. Write 2499 in Roman numerals and explain the grouping. 3

OR

Write 1666 in Roman numerals and explain the grouping. 3

18. Write 3888 in Roman numerals and explain the grouping. 3

OR

Write 949 in Roman numerals and explain the grouping. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Why did early humans need to count? 5
(ii) Describe how pebbles or sticks help in counting.
(iii) What is the limitation of using letters of the alphabet for counting?

OR

(i) Convert 1994 to Roman numerals. 5
(ii) Convert MCMXLIV to decimal.

20. (i) Convert 111010_2 to decimal. 5
(ii) Convert 58 to binary.
(iii) Add $1011_2 + 110_2$.

OR

Case study: A clock tower shows time using base-60 minutes and seconds, a tradition traced to Mesopotamia. 5
(i) How many seconds are in 3 hours?
(ii) Why is 60 convenient?
(iii) Write 3 h 25 min in minutes.

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND